

Motor Report

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I. INTRODUCTION

This paper focuses on modelling the dynamic behaviour of a brushed MC33926PNB DC motor and designing an adequate controller that relays the input from one motor to an identical equivalent. These two motors are implemented in a Segway that balances on two wheels, therefore some performance requirements must be preset. Firstly, the system must present a stable behaviour, with a fast response time, and limited overshoot. More specifically, a maximum 10% overshoot and a settling time of 0.8 seconds or less as derived from the suggested maximums outlined in the Lab Manual (1). Lastly, the designed controller should not exceed a steady state error of 5%. In order to achieve this level of control, first a model and corresponding parameters for each of the components will be found and verified against experimental data.

II. ANALYSIS

A. Segway motor

The modelled system represents an electric motor, which drives a wheel that is connected to the shaft of the motor. The model consists of three main parts;

- The electrical power supply
- The electric motor
- The shaft and wheel

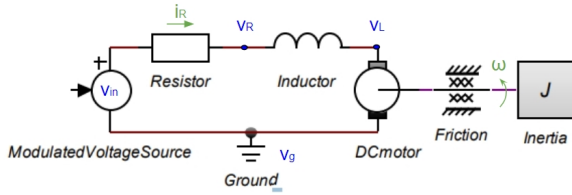


Fig. 1: Iconic model of motor, 20-sim

The shaft and wheel are described by the inertia block J , which experiences mechanical friction b . The electric motor is modelled as a gyrator with a motor constant k_r , to model the loss and non-ideal behaviour of the motor a resistor R and inductor L are connected to an input voltage V_{in} . Upon inspection of the model, it can be seen that there are two energy-storing elements present, namely an inductor and inertia, which are connected to the gyrator. The gyrator's behaviour is described by the following two equations, where τ is mechanical torque, i_R is an electrical current, ω angular velocity, and V_g is the voltage over the gyrator.

$$e_2 = k_r \cdot f_1 = k_r \cdot i_R = \tau \quad (1)$$

$$e_1 = k_r \cdot f_2 = k_r \cdot \omega = V_g \quad (2)$$

Conducting an analysis of the mechanical side of the model with equation 1 results in the following relation;

$$\tau = J\dot{\omega} + b\omega = k_r \cdot i_R \quad (3)$$

On the other hand, Kirchhoff's voltage law and equation 2 describe the electrical network as follows;

$$V_{in} = V_l + V_R + V_g \quad (4)$$

$$V_{in} = L \frac{di_R}{dt} + R \cdot i_R + k_r \cdot \omega \quad (5)$$

Combining equations 5 and 3 using the common current variable i_R , produces the input-output equation that described the behaviour of the system. Consider equation 6

$$V_{in} = \underbrace{\left(\frac{JL}{k_r}\right)}_A \ddot{\theta} + \underbrace{\left(\frac{JR}{k_r} + \frac{bL}{k_r}\right)}_B \dot{\theta} + \underbrace{\left(\frac{bR}{k_r} + k_r\right)}_C \dot{\theta} \quad (6)$$

From equation 6, it can be seen that there are five parameters that need to be determined.

This model of the motor, although being a good analogue, still makes assumptions about the behaviour of the motor, namely, that the motor constant is unchanging for a range of speeds and applied torques. Given that the motor will have a fairly small operating range (with low torques and speeds needed only for self-balancing) this is likely to be negligible given a robust controller.

B. Segway controller

Before considering the control of the system, the model is converted to a bond graph representation shown in 2.

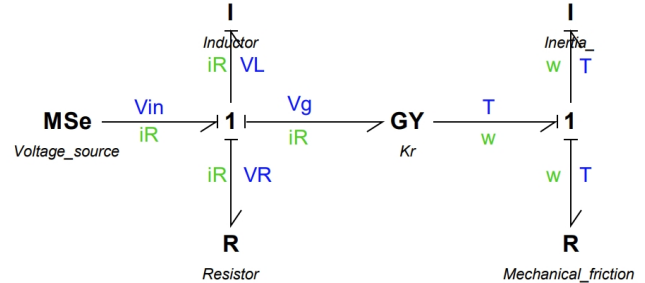


Fig. 2: Bond graph of motor, 20-sim

Furthermore, the system's transfer function is rewritten using the representation shown in equation 6 from input voltage to output angle. Consider 7, where parameters A , B and C are indicated in equation 6.

$$G(s) = \frac{U(s)}{Y(s)} = \frac{1}{As^3 + Bs^2 + Cs} \quad (7)$$

From the transfer function, it can be seen that the system will experience second-order behaviour. Unlike the first-order system, a second-order system does not directly converge to

a steady state but can oscillate. This introduces overshoot M_p and settling time, t_s . As outlined by the project manual (1), an overshoot larger than 10% and the largest time constant in the response to be below 0.2 s are sufficient preliminary design constraints for a suitable controller. Using the desired overshoot limit, the maximum damping factor ζ is determined as follows;

$$M_{p\%} = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 10\% \\ \zeta = 0.512$$

The settling time, on the other hand, is used to determine the required position of the poles $p_{1,2} = -\sigma \pm j\omega_d$, for which $t_{s2\%} = 4\tau$, and $n = 2$ according to (2);

$$t_{sn\%} \leq \frac{1}{\sigma} \ln \frac{100}{n} \leq 4 * 0.2 \\ \sigma \leq 4.89 \\ \omega_d = \sigma \frac{\sqrt{1-\zeta^2}}{\zeta} = 10.73 \\ p_{1,2} \leq -4.89 \pm 10.73j \quad (8)$$

The established poles describe the boundaries on the complex plane for which the controller's poles must lie within.

III. METHODOLOGY

A. Internal resistance (R)

The internal resistance R is calculated using the current and voltage over the motor. The system must be brought to a steady state, so that the current through resistance is remains at a constant value. Additionally, in accordance to Kirchhoff's current law, the current through the inductor does not change instantaneously, meaning the inductor behaves like a short circuit. Consider equation 9.

$$R = \frac{V_{in}}{i_R} \quad (9)$$

The current is measured for the maximum input voltage of 24V. Furthermore to achieve the steady state, the motor axis is physically clamped, to prevent it from rotating.

B. Inductance (L)

The inductance of the inductor can be determined by considering its discharging process. An exponential decay described in equation 10, is the expected behaviour, as the inductor provides current to the load.

$$i_R = \frac{V_{in}}{R} \left(e^{-\frac{R}{L}t} \right) \quad (10)$$

The decay is considered after a two-second-long pulse of 24V, by measuring the discharge in current i_R , with time. Taking the time constant of this decay τ , allows for the computation of the inductance using the relation $\tau = \frac{R}{L}$, shown in equation 10.

C. Motor constant (K_r)

The motor constant defines the behaviour of motor. It is equal to the gyrator ratio k_r of the gyrator, as described in equations 1 and 2. The torque of the motor is determined by attaching the provided lever 14cm of length l to the motor axis, and positioning it above the provided scale $\pm 0.01g$. The scale reading is multiplied by the gravitational constant to get the exerted force F . Consider the following equation, where angle θ is the angle between the lever and the scale.

$$\tau = \sin(\theta) \cdot l \cdot F \quad (11)$$

To determine the motor constant k_r , a current torque graph is plotted for a range of input voltages, where the constant is the reciprocal of the slope, as described by eh relation in equation 1.

D. Mechanical friction (b)

Mechanical friction can be determined by once again putting the system into its steady state, which brings the time derivatives in equation 6 to zero. The steady-state relation can then be rewritten into the following;

$$b = \frac{k_r}{R} \left(\frac{V_{in}}{\omega} \cdot k_r \right) \quad (12)$$

Angular velocity ω is still an unknown variable, which can be computed by measuring and plotting the change in angle over time. The slope of this linear relation is the angular velocity. Afterwards, it is used to solve for mechanical friction.

E. Inertia (J)

The start-up and cutoff transient velocity behaviour of the motor-shaft-wheel system can be analysed in order to measure the inertia of the wheel. Given a sample disc of known inertia, this value can then be calibrated for a more precise outcome. Using the general differential equation for the mechanical side of the system 3, the following relation is derived with b being the mechanical friction constant with viscous air resistance being neglected;

$$e^{-\frac{b}{J}t} \Rightarrow \tau = \frac{J}{b} \quad (13)$$

The method of using the start-up and cutoff transients comes with the assumption that the reverse EMF from the charge or discharge of the inductor can be neglected. Assuming near-identical steady states, the cutoff transient is likely to be more reliable than the start-up due to potential variations in input voltage not making the input an ideal step response. However, both will be used to find the time constant of the system for a certain inertia for redundancy and comparison and to verify this claim.

F. Controllers

Four controllers have been considered for this application, including pole placement, Phase lead-lag, feedforward, feedback, and a combination of them. The advantage of the pole placement method is the ability to use of the poles determined in equation 8, however, designing a phase lead-lag controller is preferable, since it produces a biproper controller, meaning, the number of poles is equal to the number of zeros. Furthermore, it is implementable, has a feed-through term, a biproper inverse and, a non-zero and finite high-frequency gain (3). For that reason, the phase lead-lag controller described

by equations 14, have been chosen over the pole placement method.

$$C(s) = \frac{s - z}{s - p} \quad (14)$$

For a lead controller, the pole is larger than the zero $z \leq p$, and vice versa for lag, controller $z \geq p$. In addition feed-forward control is implemented by using the product of the reference angle θ and the reciprocal of the plant, $\frac{1}{G(s)}$. However, poor accuracy is expected due to the low magnitude of the inertia.

IV. RESULTS

The summary of the found parameters is given below

TABLE I: Summary of found parameter values and units

Parameters	Units
R	Ω
L	H
k_r	W
B	N
J_{wheel}	kgm^2

A. Internal resistance (R)

The blocked steady state current yielded an average current of $1.20Amps$, alongside the input voltage of $24Volts$, the total internal resistance was found to be 19.94Ω

B. Inductance (L)

By applying a pulse to the same blocked state as for the resistance measurement, the following discharge curve was found as seen in figure 3 below, with the value of the time constant being interpolated from an average of three measurements;

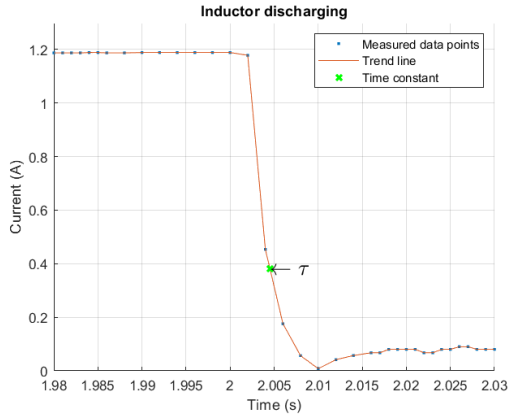


Fig. 3: Inductor discharging from a 24V input

The average value was found to be $L = 0.0906H$. The coefficient of determination R^2 is large enough to neglect the error in inductance L .

C. Motor constant (K_r)

The torque current plot as shown in figure 4, yielded the reciprocal slope equal to the motor constant of $k_r = 0.0513$;

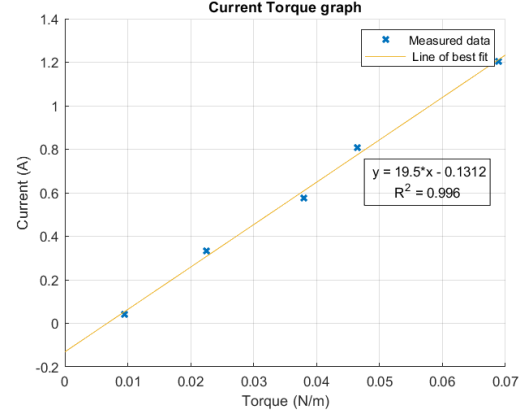


Fig. 4: Current torque graph $i_R \cdot K_r = \tau$

D. Mechanical friction (b)

By using a plot equivalent to that of figure 5 except with nothing mounted on the shaft to remove the effect of air resistance, the steady state value was used to find a mechanical friction of $b = 1.02 \cdot 10^{-4}$. Similarly to figure 5, the data was filtered using a frequency-domain low pass filter to eliminate most high frequency noise from the signal.

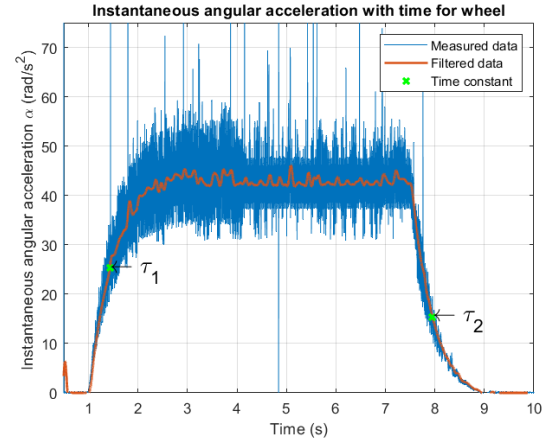


Fig. 5: Instantaneous angular acceleration with time for wheel

E. Inertia (J)

Due to the extremely fast transients for an empty shaft, this inertia was neglected with it being assumed to be a part of the calibration disk inertia. The τ and the resulting inertia for the disc and wheel are as follows in the table II below:

TABLE II: Measured time constants and inertia values

Inertial Body	(τ) Rise / Fall	Expected Inertia	Found Inertia
Disc	0.501/0.503	$4.69 \cdot 10^{-5}$	$5.11 \cdot 10^{-5}$
Wheel	0.418/0.423	<i>Unknown</i>	$4.26 \cdot 10^{-5}$

Given that the discrepancy between the two time constants for both bodies are very small, it can be concluded that the assumption highlighted earlier had a small affect on the final time constant value. Given the difference in the value for the inertia of the calibrated disc, the constant of the wheel will also be offset proportional to the difference for the disc based on validation data, this yields the following plot in figure 6 of all three data sets below with a difference in inertia of 16%. This plot also verifies that the found inertia is correct as seen by the matching startup and cutoff transient, with a slight offset in the start time of the cutoff due to a mismatch in pulse length, yet still valid due to an identical curve.

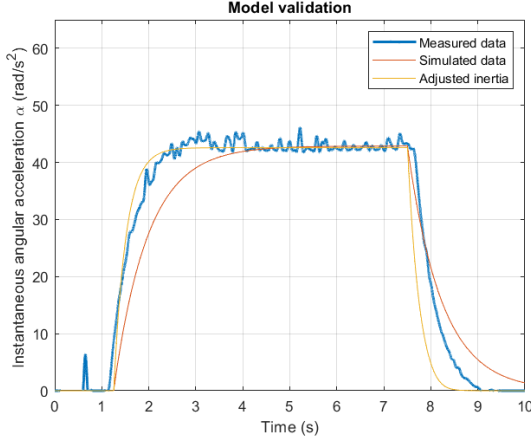


Fig. 6: Model validation

F. Parameter validation

For each parameter, the effects of an incorrect value on the simulated vs. real data will be discussed respectively. The resistance determines the amount of current in the circuit, so a higher resistance for the same voltage would decrease the steady state value, which is not the case in figure 6, the same discrepancy would be seen for an incorrect motor constant aswell. A higher value for either of the P-type elements (inductor and rotating wheel) would increase the time constant of the system as seen in the unadjusted data in figure 6, the new data for the adjusted error in the inertia has fixed this behaviour, and the model behaves similarly to the system, with both elements potentially having a slightly small value, but within the error margins of matching of the pulse lengths. A higher inductance would slow down the acceleration of the mass due it also being a P-type element same as the inertia of the wheel. An excessive inertia, as observed in figure 6, would result in

G. Controllers validation

Using the preset location requirement of the poles on the complex plane, from equation 8, the determined phase lead and lag controller poles and zeros are in table III.

TABLE III: Poles and zeros of Phase lead and lag controllers

	Pole p	Zero z
C_{Lead}	20.21	1.23
C_{Lag}	12.31	1.01

The root locus for these poles is shown in fig 7, where two of the poles are not visible due to a very negative real value.

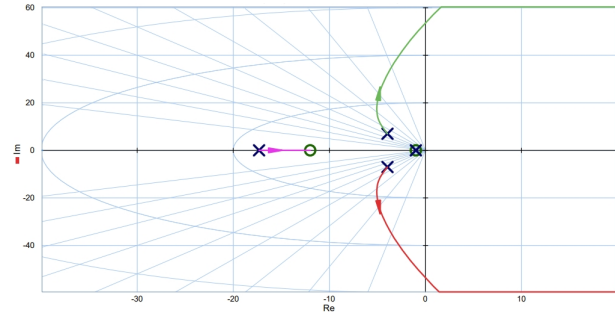


Fig. 7: Pole assignment in root locus

As indicated in fig 7, the poles are situated in $-5.21 \pm 9.87j$ meaning the controller falls within the desired location. To validate the model and verify the steady state error, the input angle has been plotted against the referenced angle for a sine and ramp input. Consider fig. 8, showing that the designed controller falls within the parameters.

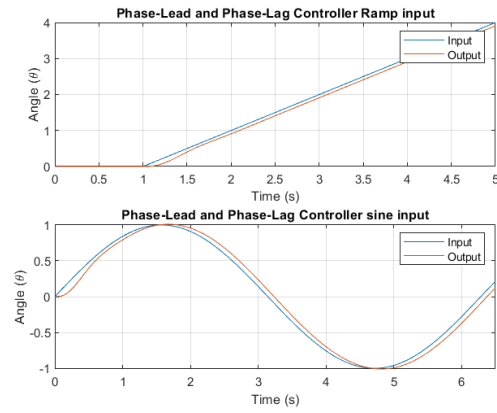


Fig. 8: Steady state error for Phase lead-lag controller

Lastly, to validate the controller, a system was subject to a step response to verify if the requirements are met. Consider fig. 9.

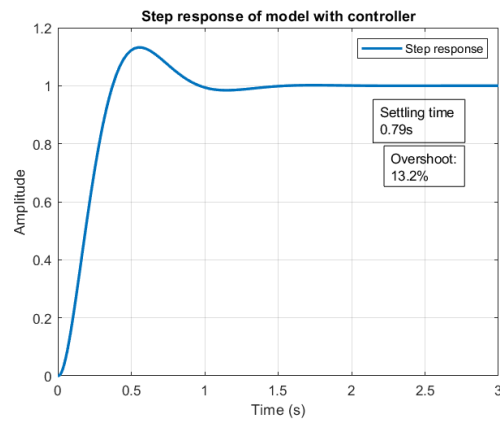


Fig. 9: Step response of system

As indicated in the figure, the settling time t_s , meet the requirement of eight seconds or less, however the overshoot, is larger than expected at 13.2%, which is a 3.2% error.

V. CONCLUSION

Regardless of issues encountered with transmitting Segway data to a laptop, a successful model for the motor was developed, however, some further issues were encountered when developing the control system. The parameters were all determined to a sufficient degree of accuracy, which is reflected in the precise behaviour of the simulation when compared to the experimental step input. The controller design was up to specification for the steady-state behaviour, with a less than 5% steady-state error to a ramp and sinusoidal input, which also served as an additional verification step for the determined model parameters. A controller capable of a step response up to our defined specification was not fully developed, with the defined maximum overshoot of 10% of the step value, and a 2% settling time of 0.8 seconds, equalling a maximum time constant of 0.2 seconds. The final developed controller achieved an overshoot of 13.2% and 0.79 settling time.

VI. DISCUSSION

During the experiment, it has been decided to exclude a feedforward controller, because it had very little to no impact on the system's response. Its inaccuracy, is most likely due to the predicted reason of a small inertia value. This was confirmed during the testing of the controller, as no visible difference was spotted. On the other hand, the phase lead-lag controller, didn't meet the overshoot requirement. Since the overshoot is directly impacted by the damping ratio, ζ , the source of the error points towards, term B , in the transfer function. Due to the reoccurrence of the parameter values, in the other coefficients, it is unlikely that the error lies within a parameter measurement. Instead, an design mistake in the controller is more probable. To address this issue, the design of the controller will be revisited following the completion of this report.

REFERENCES

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